AHMED BEN SALAH

ICT Engineering Student | AI Engineering & Machine Learning | Cybersecurity

ICT Engineering Student | Machine Learning | Adversarial ML & AI Security

ICT Engineering Student | Offensive Security & CTF | Security of AI Systems

ICT Engineering Student | Software Engineering | Backend, AI Integration & Competitive Programming

ICT Engineering Student | Autonomous Robotics | Embedded Control & CAD

ahmedbensalah.professional@gmail.com ✦ +216 94 466 757 ✦ Tunis, Tunisia ✦ linkedin.com/in/ahmedbensalahh
github.com/Ahmedd-Ben-Salah ✦ ahmedd-ben-salah.github.io ✦ codeforces.com/profile/A_ZeroOne

PROFESSIONAL SUMMARY

ICT Engineering student at INSAT (2024–2029), Networks & Telecommunications track, with a self-directed focus on AI engineering, machine learning and cybersecurity. Builds and ships production AI systems — multi-agent RAG pipelines, LLM-driven network automation and autonomous robotics control. Two-time robotics competition podium finisher (NXP Cup, IEEE Micromouse), with certifications from Stanford/DeepLearning.AI, Harvard and Cisco. Particular interest in the intersection of AI and security, where he has published a from-the-paper adversarial attack on a neural network classifier.

ICT Engineering student at INSAT (2024–2029) with a self-directed focus on machine learning and the security of machine-learning systems. Built and shipped production AI systems — multi-agent RAG pipelines, LLM-driven automation and autonomous robotics control — and has published an FGSM adversarial attack that drops a 97.5%-accurate classifier to 6.5% with perturbations invisible to the eye. Stanford/DeepLearning.AI certified, two-time national robotics podium finisher.

ICT Engineering student at INSAT (2024–2029), Networks & Telecommunications track, focused on offensive security and the security of AI systems. Ran a security and QA review of a live SaaS platform, places in national and regional CTFs, and has published a working adversarial attack that breaks a neural network classifier with imperceptible input perturbations. CCNA certified, CTF competitor on PicoCTF and TryHackMe.

ICT Engineering student at INSAT (2024–2029) who ships production software: social-media publishing integrations, authentication with one-time codes, automated document generation and LLM provider routing, all in a live multi-agent SaaS platform. Competitive programmer training toward TCPC/ACPC, two-time national robotics podium finisher.

ICT Engineering student at INSAT (2024–2029) with hands-on autonomous-robotics experience: CAD design, real-time PID control and maze-solving algorithms on real hardware. Two-time national robotics podium finisher (IEEE Micromouse, NXP Cup), Certified SOLIDWORKS Associate, and an AI engineer who ships production software.

EDUCATION

National Institute of Applied Science and Technology (INSAT)

Engineering Degree in Information and Communication Technologies (ICT)

Track: Networks & Telecommunications · Self-directed focus: AI Engineering, Machine Learning, Cybersecurity

Tunis, Tunisia · 2024 – 2029 (expected)

- 5-year engineering programme covering networking, telecommunications, software engineering, AI, machine learning, data science and embedded systems.

EXPERIENCE

Welyne · *ai-commandos.com*

AI & Automation Engineer · July 2026 – September 2026 · Tunis, Tunisia

- Integrated LinkedIn, Facebook and Instagram publishing into the production platform, verified end to end with real posts.
- Implemented per-workspace selection of the language model and provider, and email verification with one-time codes at signup.
- Built PowerPoint deck generation and reworked the automated PDF reports produced by the platform's market-intelligence agent.
- Carried out a security and QA review of the deployed application, reporting issues such as authentication tokens passed in URLs and access checks that failed open.

Welyne · *ai-commandos.com*

AI & Automation Intern · April 2026 – July 2026 · Tunis, Tunisia · On-site

- Contributed to AI Commandos, a production platform of nine specialized AI business agents, working primarily on **Hunter** (prospecting, follow-up and lead reactivation) and **Nova** (marketing and social media).
- Built LinkedIn prospecting automation for Hunter: profile scraping of high-value targets (CEOs, C-suite executives), automated multi-channel follow-up sequences and LLM-personalized outreach at scale.
- Worked on Nova's automated social campaign features — content generation, editorial calendar planning and campaign scheduling.

PROJECTS

Adversarial Attacks on a Neural Network Classifier (FGSM)

Python · TensorFlow/Keras · Adversarial Machine Learning · NumPy · Matplotlib · pytest
github.com/Ahmedd-Ben-Salah/fgsm-mnist · October 2026

- Implemented the Fast Gradient Sign Method directly from Goodfellow et al. (ICLR 2015), differentiating the loss with respect to the **input image** rather than the weights.
- Collapsed a classifier from **97.5% to 6.5% accuracy** at a perturbation of ε = 0.1, and to 1.5% — below the 10% of random guessing — at ε = 0.15, while the attacked images remained visually indistinguishable from the originals.
- Measured the full accuracy-versus-perturbation curve across nine attack strengths, establishing that the bulk of the failure occurs while the perturbation is still imperceptible.
- Published as a reproducible public repository with an isolated attack module, a test suite covering the attack's invariants, and documented results.

IEEE Micromouse — 3rd Place, First Micromouse Competition in Tunisia

C++ · Flood-Fill Pathfinding · PID Control · CAD · ToF Sensors · Encoders · Gyroscope · Embedded Systems
National Robotics Week, INSAT · September 2026 · Team of 5

- Placed 3rd of 7 teams with an autonomous robot that maps and solves a 16×16 maze.
- Owned the full CAD design and all of the software, including PID motion control and pathfinding.
- Implemented flood-fill maze solving, updating the distance map as walls were discovered by time-of-flight sensors.
- Built closed-loop motion on N20 motors with encoders and a gyroscope, with an on-board OLED menu for tuning parameters on the track.

NXP Cup — 3rd Place, National Autonomous Car Competition

C++ · Teensy Microcontroller · PID Control · SolidWorks · Embedded Systems · IEEE INSAT RAS

- Placed 3rd of 16 teams nationally as the youngest team in the competition, after the main Teensy board short-circuited 60 minutes before the race with no test runs available on the replacement hardware.
- Owned a hybrid software/hardware role: designed chassis components in SolidWorks for structural precision while writing the real-time PID control software for high-speed line tracking.
- Flashed firmware onto borrowed hardware with zero calibration time; the robot completed the full track and stopped 5 cm from the target cube on the second lap.

CiscoLabAI — AI-Powered Network Automation Workflow

Python · LLM Integration · Prompt Engineering · Cisco Packet Tracer · API Integration · NetDevOps

- Built an AI automation workflow that reduced Cisco Packet Tracer lab configuration from 70 minutes of manual CLI work to under 3 minutes — a 95% reduction in manual effort.
- Automated full dual-stack (IPv4 + IPv6) addressing, device hardening and advanced routing by integrating LLM-driven intent parsing with configuration-generation pipelines.
- Applied intent-based automation principles: the tool interprets architecture-level goals and generates and deploys the correct configuration sequences autonomously.

Dalanda — Multi-Agent AI Automation System

Python · LangChain · FAISS · React · Multi-Agent AI · RAG · Prompt Engineering

- Built a multi-modal, multi-agent RAG system that ingests images, audio and documents, routing each modality to a specialized model for extraction and structuring.
- Implemented an agent verification pipeline that cross-checks generated responses against source data, reducing hallucinations without human review.
- Built API-connected workflows enabling semantic FAISS vector search with filtering, grounding LLM output in verified source data.
- Delivered a React interface with voice input, letting non-technical users drive the pipeline in natural language.

TECHNICAL SKILLS

AI Engineering	LLM Integration, Prompt Engineering, Multi-Agent Systems, RAG Pipelines, LangChain, FAISS Vector Search, Autonomous Agents, AI Workflow Automation
AI / ML	Machine Learning, Neural Networks, TensorFlow, Supervised Learning, Linear & Logistic Regression, Decision Trees, Random Forests, XGBoost, Gradient Descent, Feature Engineering, NumPy, Pandas, scikit-learn
Networking	Network Architecture, Routing & Switching, IPv4/IPv6 Dual-Stack, Protocol Analysis, CCNA Networking, NetDevOps
Cybersecurity	Network Security, Penetration Testing, Ethical Hacking, Wireshark, tcpdump, Nmap, Linux (Kali), TryHackMe, HackTheBox
Automation	Web Scraping, Workflow Automation, API Integration, Pipeline Design, Intent-Based Automation
Languages	Python, C++, C, Java, JavaScript, R, SQL, Bash/Shell Scripting
Web & APIs	Full-Stack Development, React, Node.js, REST APIs, Flask, HTML/CSS, Databases (SQL/NoSQL)

Robotics & Embedded	C++, Real-Time PID Control, Flood-Fill Pathfinding, SolidWorks CAD, ToF Sensors, Encoders, Gyroscope, Teensy, Arduino, Embedded Systems
Tools & DevOps	Git, GitHub, Docker (basic), VS Code, Linux (Arch), Arduino, SolidWorks

COMPETITIVE PROGRAMMING & SECURITY PRACTICE

- **Codeforces** — active competitor (A_ZeroOne); training toward TCPC and ACPC. Focus: binary search, greedy, sorting, prefix sums, two pointers.
- **TryHackMe** — tryhackme.com/p/AZeroOne — Cyber Security 101 path in progress: Linux fundamentals, Windows & Active Directory, command line, networking (Wireshark, tcpdump, Nmap).
- **PicoCTF & OverTheWire** — capture-the-flag challenges across forensics, general skills and Linux privilege exercises.

CERTIFICATIONS

Advanced Learning Algorithms	Stanford / DeepLearning.AI	*Neural networks (TensorFlow), bias/variance, decision trees, XGBoost*
Supervised Machine Learning: Regression and Classification	Stanford / DeepLearning.AI	*Regression, classification, gradient descent, regularization*
CS50: Introduction to Computer Science	Harvard / edX	*Algorithms, data structures, C, Python, SQL*
Applied Calculus	Harvard (Verified)	*Applied mathematics and quantitative reasoning*
CCNA — Introduction to Networks	Cisco	*Routing, switching, networking fundamentals*
Certified SOLIDWORKS Associate (CSWA)	Dassault Systèmes	*CAD, 3D modeling, mechanical design*

LANGUAGES

Arabic (Native) · **English** (Professional) · **French** (Professional) · **Spanish** (Beginner)